

Lab 10 Worksheet — Network Cabling —

Termination, Testing and Tool Selection

Name: ____ Date: ____

Exam objective: Terminate twisted-pair cable to T568B, test it end to end, and select the correct tool for each cabling task (Core 1 objectives 2.3 and 3.1).

Record as you go

Step	What you did	What you observed
1	Write out the T568B colour order from memory before you begin: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.	
2	Strip about 25 mm of the outer jacket with the cable stripper, taking care not to nick the insulation of the individual conductors.	
3	Untwist and straighten the four pairs, then arrange all eight conductors flat in the T568B order you wrote down.	
4	Trim the conductors square to about 12 mm so they will all reach the end of the plug while the jacket still enters the strain relief.	
5	Insert the conductors fully into the RJ45 plug, confirm through the clear plastic that each wire reaches the end and the order is still correct, then crimp firmly.	
6	Terminate the far end to T568B as well, which makes this a straight-through patch cable rather than a crossover.	
7	Test the cable with the cable tester and confirm the lights advance 1 through 8 in sequence on both the main unit and the remote.	

Step	What you did	What you observed
8	Deliberately make a second cable with two conductors swapped, test it, and record how the tester reports the crossed pair.	
9	Make a third cable with one conductor not fully seated, test it, and record how the tester reports the open circuit.	
10	Use the toner probe on a bundle: attach the tone generator to one known end and use the probe to find the matching far end.	
11	Terminate a cable to a patch panel or keystone jack with the punch-down tool, noting that the blade trims the excess conductor as it seats.	
12	Build the tool selection guide: crimper terminates plugs, stripper removes the jacket, cable tester verifies the pinout, toner probe identifies a cable in a bundle, punch-down seats wires in a block, loopback plug tests a port, and a Wi-Fi analyser shows channel congestion.	

Verification

Success criterion: Your good cable passes the tester with all eight lights in sequence on both ends; the two faulty cables produce distinctly different tester results that you have recorded; and the tool guide matches all seven tools to the specific question each answers.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?

